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# PIZZA

A Delicious Journey

# Pizza Sales Analysis Project

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## Overview

This project explores pizza sales data using MySQL, analyzing over 20,000 customer orders across 32 different pizza types. The goal was to uncover meaningful business insights by applying advanced SQL techniques.

## Data Model

The analysis was based on four core tables:

orders – captures order-level information (order ID, date, time).

order\_details – contains line items for each order (pizza ID, quantity).

pizzas – stores pizza size and price details.

pizza\_types – defines categories and names of pizzas.

## Methodology

To extract insights, I applied:

### Joins:

Two-table joins (e.g., linking orders with order\_details).

Four-table joins (combining all tables for complete sales analysis).

Window functions: Used for ranking pizzas by popularity, calculating cumulative sales, and identifying top contributors.

CTEs (Common Table Expressions): Simplified complex queries, making the analysis modular and easier to interpret.



# Tables Used

- 1 . Oder\_details (Order\_details\_id , Order\_id , Pizza\_id , Quantity)
- 2 . Orders (Order\_id , Order\_Date , Order\_Time)
- 3 . Pizzza\_Types (Pizza\_type\_id , Name , Category , Ingredients)
- 4 . Pizzas (Pizza\_id , Pizza\_Type\_id , Size , Price

# Retrieve the total number of orders placed.

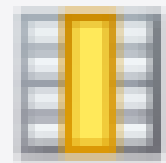
```
select count(order_id) as Total_Orders from orders;
```

| Result Grid |              |
|-------------|--------------|
|             | Total_Orders |
| ▶           | 21350        |

# Calculate the total revenue generated from pizza sales.

```
SELECT
    ROUND(SUM(order_details.quantity * pizzas.price),
          2) AS Total_Sales
FROM
    order_details
    JOIN
    pizzas ON pizzas.pizza_id = order_details.pizza_id;
```

Result Grid



|   | Total_Sales |
|---|-------------|
| ▶ | 817860.05   |

# Identify the highest-priced pizza.

```
SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
    JOIN
        pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY price DESC
LIMIT 1;
```

| Result Grid |                 |       | Filter Rows |  |
|-------------|-----------------|-------|-------------|--|
|             | name            | price |             |  |
| ▶           | The Greek Pizza | 35.95 |             |  |

# Identify the most common pizza size ordered.

```
SELECT
    pizzas.size,
    COUNT(order_details.order_details_id) AS Total_Count
FROM
    pizzas
    JOIN
    order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size
ORDER BY Total_count DESC;
```

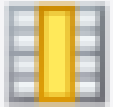
| Result Grid |      |             | File |
|-------------|------|-------------|------|
|             | size | Total_Count |      |
| ▶           | L    | 18526       |      |
|             | M    | 15385       |      |
|             | S    | 14137       |      |
|             | XL   | 544         |      |
|             | XXL  | 28          |      |

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# List the top 5 most ordered pizza types along with their quantities.

```
SELECT
    pizza_types.name,
    SUM(order_details.quantity) AS Total_Quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY total_quantity DESC
LIMIT 5;
```

| Result Grid     Filter Rows: <input type="text"/> |                            |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------|
|                                                                                                                                                                                                                         | name                       | Total_Quan |
| ▶                                                                                                                                                                                                                       | The Classic Deluxe Pizza   | 2453       |
|                                                                                                                                                                                                                         | The Barbecue Chicken Pizza | 2432       |
|                                                                                                                                                                                                                         | The Hawaiian Pizza         | 2422       |
|                                                                                                                                                                                                                         | The Pepperoni Pizza        | 2418       |
|                                                                                                                                                                                                                         | The Thai Chicken Pizza     | 2371       |



# Join the necessary tables to find the total quantity of each pizza category ordered.

```
SELECT
    pizza_types.category,
    SUM(order_details.quantity) AS Quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC;
```

| Result Grid |          |          | Filter |
|-------------|----------|----------|--------|
|             | category | Quantity |        |
| ▶           | Classic  | 14888    |        |
|             | Supreme  | 11987    |        |
|             | Veggie   | 11649    |        |
|             | Chicken  | 11050    |        |

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# Determine the distribution of orders by hour of the day.

```
SELECT
    HOUR(order_time) AS Hour, COUNT(order_id) AS Order_Count
FROM
    orders
GROUP BY hour;
```

| Result Grid     Filter |      |             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------|
|                                                                                                                                                                                              | Hour | Order_Count |
| ▶                                                                                                                                                                                            | 11   | 1231        |
|                                                                                                                                                                                              | 12   | 2520        |
|                                                                                                                                                                                              | 13   | 2455        |
|                                                                                                                                                                                              | 14   | 1472        |
|                                                                                                                                                                                              | 15   | 1468        |
|                                                                                                                                                                                              | 16   | 1920        |
|                                                                                                                                                                                              | 17   | 2336        |
|                                                                                                                                                                                              | 18   | 2399        |

# Join relevant tables to find the category-wise distribution of pizzas.

```
SELECT
    category, COUNT(name)
FROM
    pizza_types
GROUP BY category;
```

| Result Grid |          |             | Filter Rows |
|-------------|----------|-------------|-------------|
|             | category | COUNT(name) |             |
| ▶           | Chicken  | 6           |             |
|             | Classic  | 8           |             |
|             | Supreme  | 9           |             |
|             | Veggie   | 9           |             |

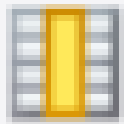
# Group the orders by date and calculate the average number of pizzas ordered per day.

```
SELECT
    ROUND(AVG(quantity), 0)
FROM
    (SELECT
        orders.order_date, SUM(order_details.quantity) AS Quantity
    FROM
        orders
    JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date) AS Order_quantity;
```

|                                                                                       |                         |                                                                                       |                                                                                       |            |
|---------------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------|
| Result Grid                                                                           |                         |  |  | Filter Row |
|                                                                                       | ROUND(AVG(quantity), 0) |                                                                                       |                                                                                       |            |
|  | 138                     |                                                                                       |                                                                                       |            |

# Determine the top 3 most ordered pizza types based on revenue.

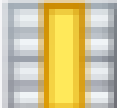
```
SELECT
    pizza_types.name,
    SUM(order_details.quantity * pizzas.price) AS Revenue
FROM
    pizza_types
    JOIN
    pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```

| Result Grid     Filter Rows: Neil Tran |                              |          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------|
|                                                                                                                                                                                                                  | name                         | Revenue  |
| ▶                                                                                                                                                                                                                | The Thai Chicken Pizza       | 43434.25 |
|                                                                                                                                                                                                                  | The Barbecue Chicken Pizza   | 42768    |
|                                                                                                                                                                                                                  | The California Chicken Pizza | 41409.5  |



# Calculate the percentage contribution of each pizza type to total revenue.

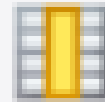
```
select pizza_types.category ,  
round(sum(order_details.quantity * pizzas.price) / (select  
round(sum(order_details.quantity * pizzas.price),2) as Total_Sales  
from order_details  
join pizzas  
on pizzas.pizza_id = order_details.pizza_id) * 100 , 2 ) as Revenue  
from pizza_types join pizzas  
on pizza_types.pizza_type_id = pizzas.pizza_type_id  
join order_details  
on order_details.pizza_id = pizzas.pizza_id  
group by pizza_types.category order by revenue desc;
```

| Result Grid |          |         |  |  | Filter |
|-------------|----------|---------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------|
|             | category | Revenue |                                                                                      |                                                                                      |        |
| ▶           | Classic  | 26.91   |                                                                                      |                                                                                      |        |
|             | Supreme  | 25.46   |                                                                                      |                                                                                      |        |
|             | Chicken  | 23.96   |                                                                                      |                                                                                      |        |
|             | Veggie   | 23.68   |                                                                                      |                                                                                      |        |

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# Analyze the cumulative revenue generated over time.

```
select order_date ,  
sum(revenue) over(order by order_date) as cum_revenue  
from  
(select orders.order_date ,  
sum(order_details.quantity * pizzas.price) as Revenue  
from order_details join pizzas  
on order_details.pizza_id = pizzas.pizza_id  
join orders  
on orders.order_id = order_details.order_id  
group by orders.order_date) as Sales;
```

| Result Grid    |            |                     |  Filter Rows: <input type="text"/> |
|----------------------------------------------------------------------------------------------------|------------|---------------------|------------------------------------------------------------------------------------------------------------------------|
|                                                                                                    | order_date | cum_revenue         |                                                                                                                        |
| ▶                                                                                                  | 2015-01-01 | 2713.85000000000004 | No Filter                                                                                                              |
|                                                                                                    | 2015-01-02 | 5445.75             |                                                                                                                        |
|                                                                                                    | 2015-01-03 | 8108.15             |                                                                                                                        |
|                                                                                                    | 2015-01-04 | 9863.6              |                                                                                                                        |
|                                                                                                    | 2015-01-05 | 11929.55            |                                                                                                                        |
|                                                                                                    | 2015-01-06 | 14358.5             |                                                                                                                        |

# Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```
select name , revenue from
(select category , name , revenue ,
rank() over (partition by category order by revenue desc) as
from
(select pizza_types.category , pizza_types.name ,
sum((order_details.quantity) * pizzas.price) as Revenue
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.category , pizza_types.name) as a) as b
where rn <= 3;
```

| Result Grid |                              |          | Filter Rows: |
|-------------|------------------------------|----------|--------------|
|             | name                         | revenue  |              |
| ▶           | The Thai Chicken Pizza       | 43434.25 |              |
|             | The Barbecue Chicken Pizza   | 42768    |              |
|             | The California Chicken Pizza | 41409.5  |              |
|             | The Classic Deluxe Pizza     | 38180.5  |              |
|             | The Hawaiian Pizza           | 32273.25 |              |